

WeB 2022 | E-Biz Workshop 2022

Is It Still a U-shape?

Empirical Study on the Dynamics of Reward-Based
Crowdfunding with Award Effects

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JaeHong Park Kyung Hee University

Part 1 ---

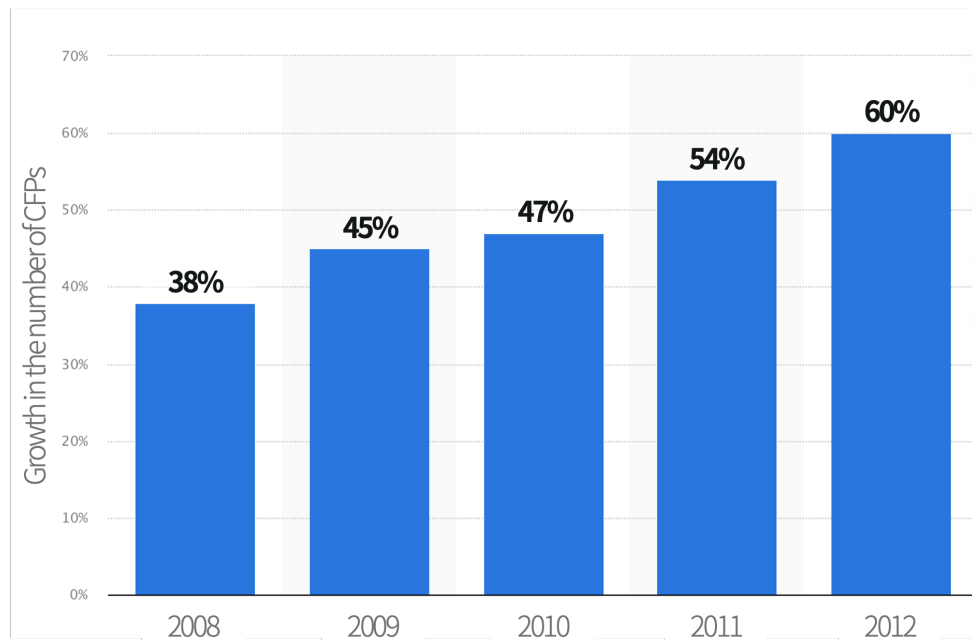
Research Background

Research Background

Crowdfunding Market

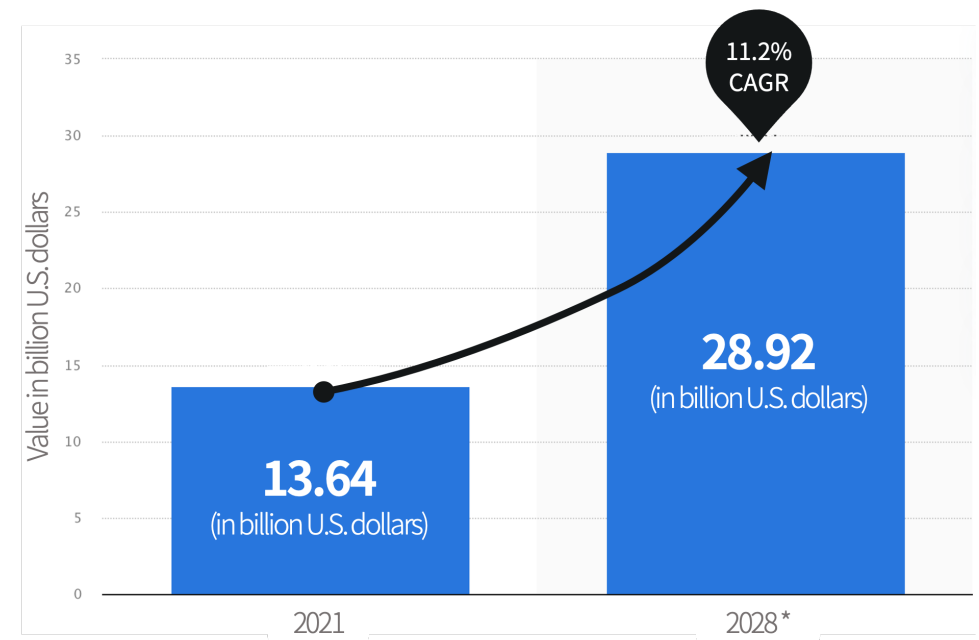
- The growth rate of crowdfunding platform has increased every year, and the growth rate reached 60 percent in 2012 (Statista 2012)
- The size of the global crowdfunding market was estimated at \$13.64 billion in 2021 and was expected to double by 2028 (Statista 2022)

Percentage growth of crowdfunding platforms worldwide



*<https://www.statista.com/statistics/251567/growth-of-crowdfunding-platforms-worldwide/>

Market size of crowdfunding worldwide

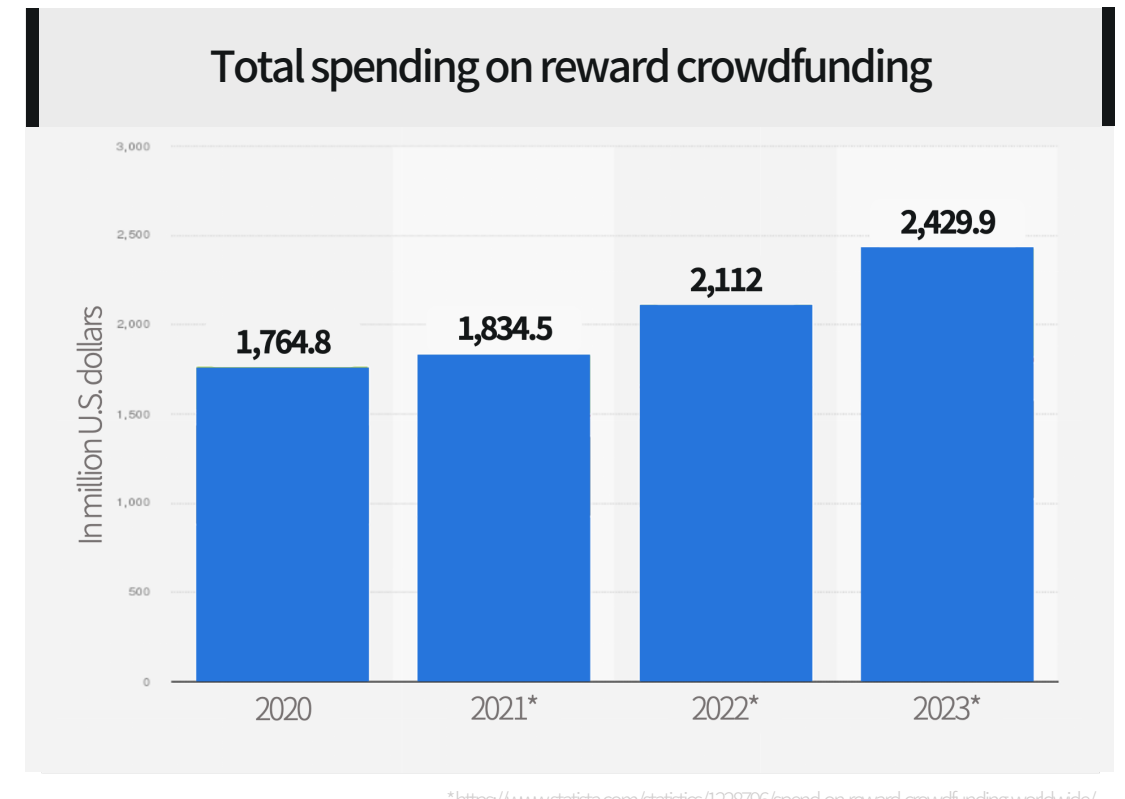
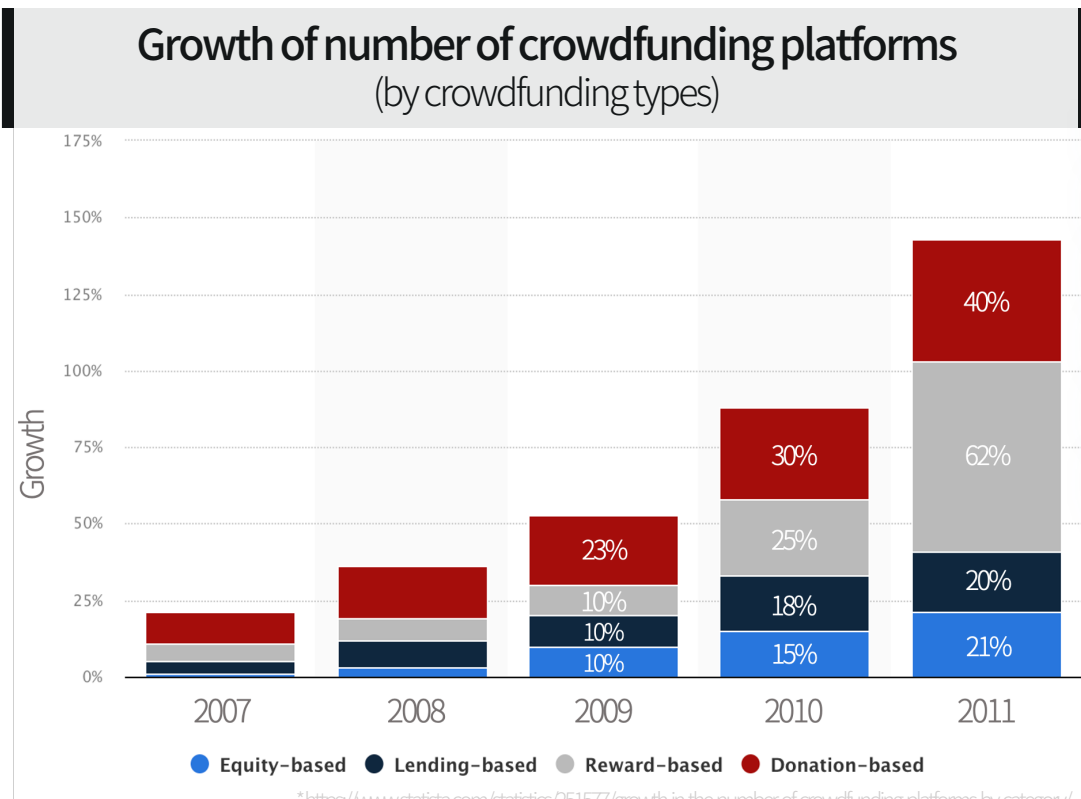


*<https://www.statista.com/statistics/1078273/global-crowdfunding-market-size/>

Research Background

Reward-Based Crowdfunding Market

- Reward-based crowdfunding market has experienced substantial growth (Statista 2012)
- Total spent money on reward crowdfunding worldwide is expected to grow approximately 700 million U.S. dollars between 2020 and 2023 (Statista 2022)

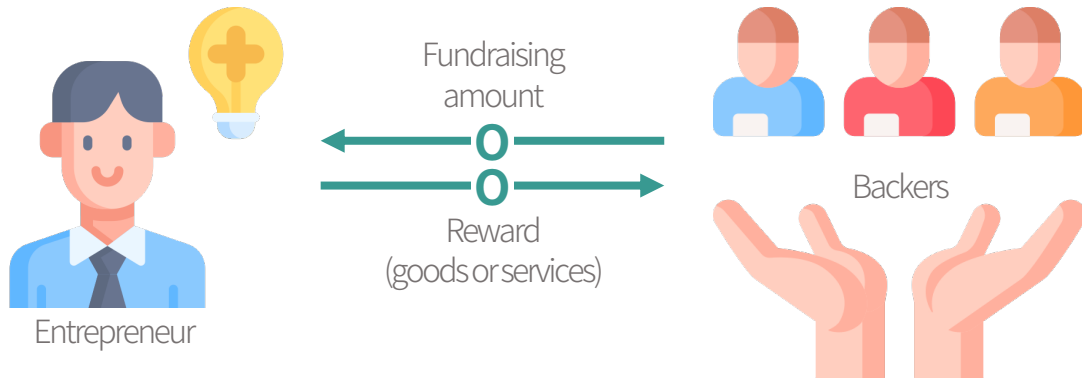


Research Background

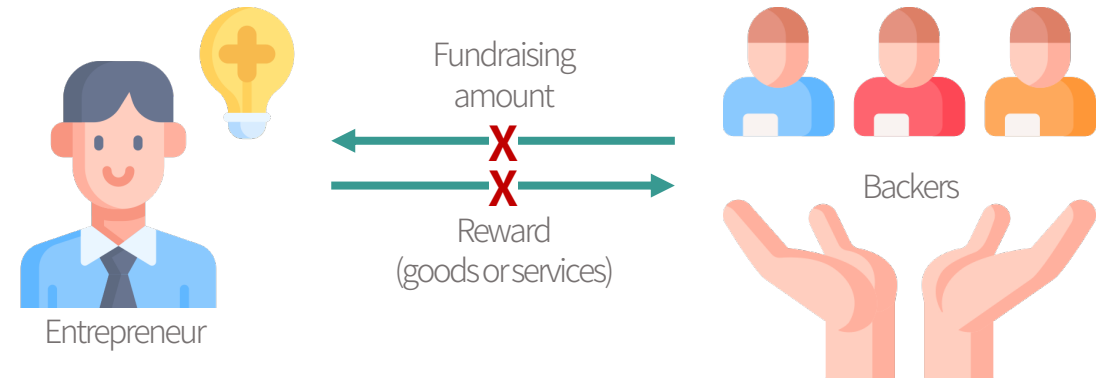
All-or-Nothing Fundraising Mechanism

- Entrepreneurs receive funding, while backers receive the reward (e.g., goods or services) only when the funding goal is achieved at the target goal amount (Li et al. 2020)
- About 59% of reward-based crowdfunding fails (Kuppuswamy and Bayus 2018)

Project achievement rate $\geq 100\%$



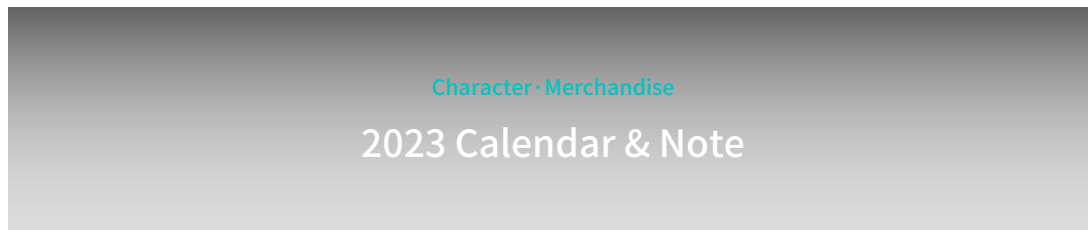
Project achievement rate $< 100\%$



Research Background

Rational Herding Behavior

- Potential backers must make a funding decision under uncertainty
- Uncertainty about crowdfunding success and reward quality can lead potential backers to observe and imitate other backers' behaviors who have already funded (Vismara 2016, Zhang and Liu 2012)



[Story](#) [Return · Policy](#) [Updates](#) [Community](#) [Supporter 2](#)



7 days to go
8% Achievement rate
\$42 Fundraised amount
2 Backers

[Pledge](#)

[3](#) [FAQ](#) [Share](#)

Hmm.. There are only two people who funded this project..

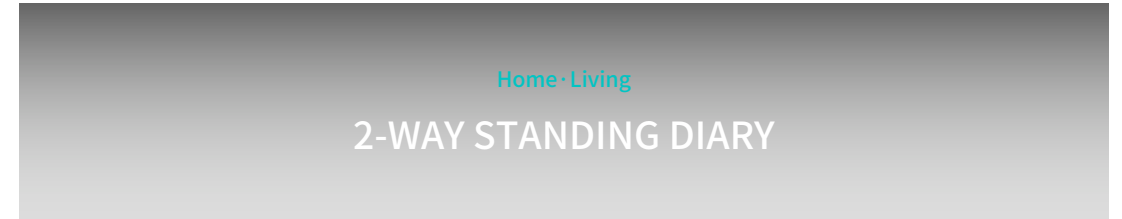


Research Background

Rational Herding Behavior

- Potential backers must make a funding decision under uncertainty
- Uncertainty about crowdfunding success and reward quality can lead potential backers to observe and imitate other backers' behaviors who have already funded (Vismara 2016, Zhang and Liu 2012)

122 people have funded this project!
I should fund it, too.



[Story](#) [Return](#) · [Policy](#) [Updates](#) [Community](#) [Supporter](#) [122](#)

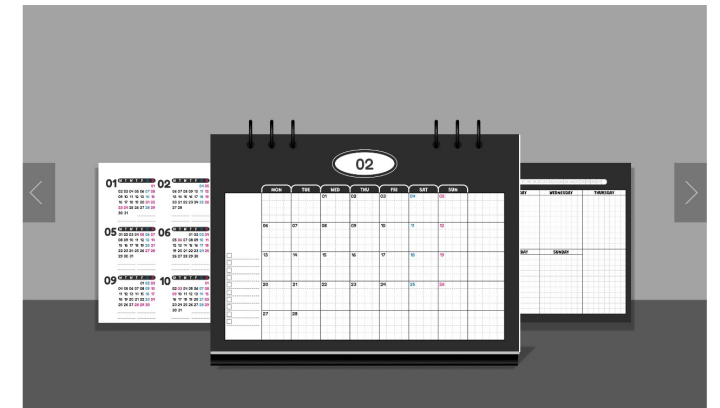
7 days to go
164% Achievement rate
\$951 Fundraised amount
122 Backers

Pledge

♥ 124

FAQ

Share



Research Background

Rational Herding Behavior

- Potential backers must make a funding decision under uncertainty
- Uncertainty about crowdfunding success and reward quality can lead potential backers to observe and imitate other backers' behaviors who have already funded (Vismara 2016, Zhang and Liu 2012)

Hmm.. There are only two people who funded this project..



122 people have funded this project!
I should fund it, too.

However,

When there is a lower level of uncertainty (e.g., award effect),
will potential backers still follow prior backers' behaviors?

Part 2 ---

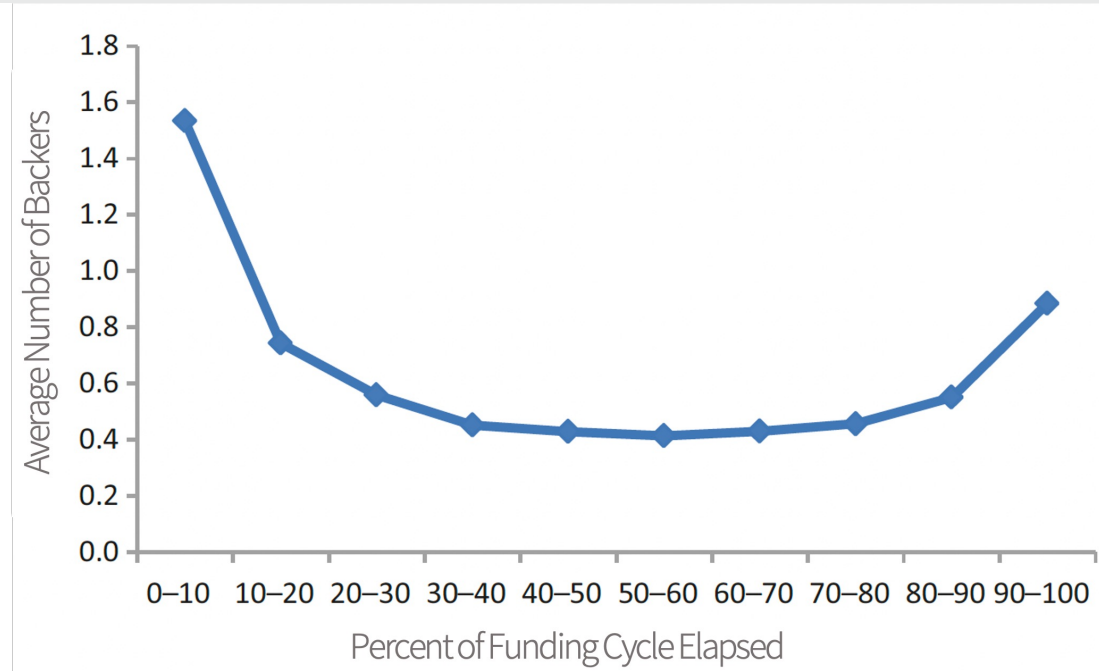
Previous Works

Previous Works

Dynamic Research of Crowdfunding

- Because potential backers tend to follow others' decisions, some prior researchers focused on the relationship between prior funding and subsequent contributions
- U-shape dynamics of crowdfunding backers' behaviors over the crowdfunding cycle (Chan et al. 2020, Kuppuswamy and Bayus 2018)

Average number of backers added to a project on any day



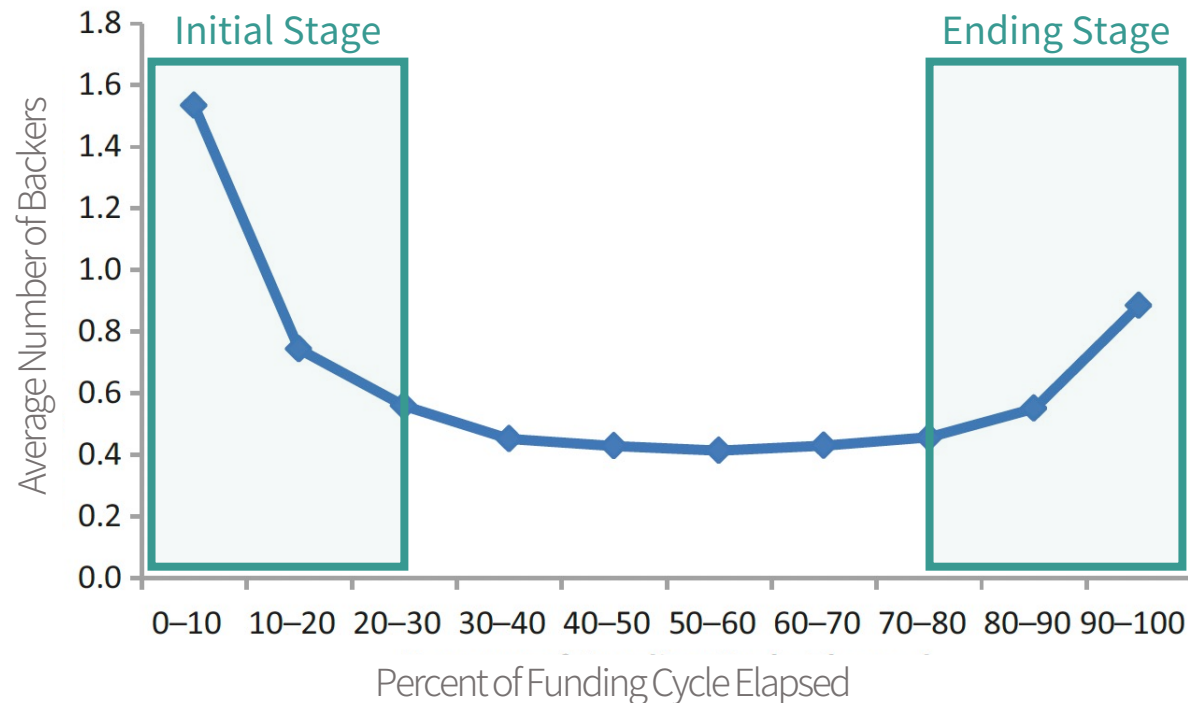
(Kuppuswamy and Bayus 2018)

Previous Works

Dynamic Research of Crowdfunding

- The funding amount decreases after the initial stage and increases at the ending stage
- Potential backers join more in the initial or ending period of the project than in the middle
(Kuppuswamy and Bayus 2018)

Average number of backers added to a project on any day



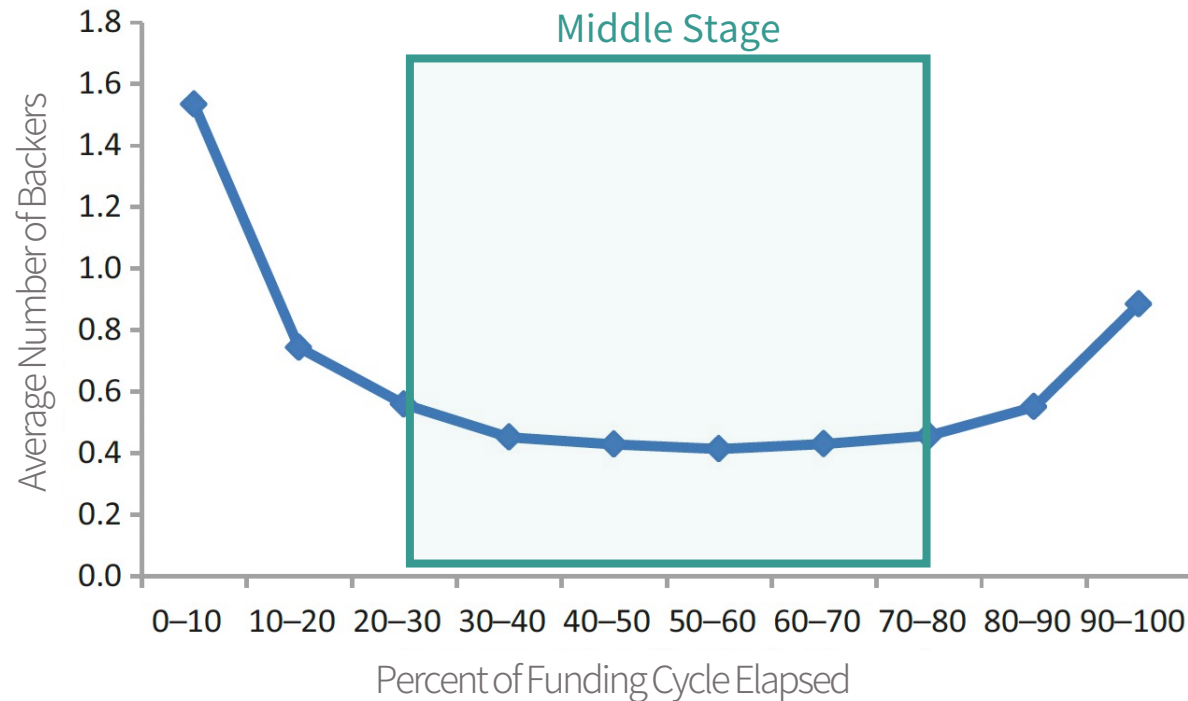
(Kuppuswamy and Bayus 2018)

Previous Works

Dynamic Research of Crowdfunding

- Additionally, fundraising is often stuck in the middle stage (Ordanini et al. 2020)
- The crowdfunding pattern over the funding cycle is U-shape regardless of success, target fundraising, and project category (Kuppuswamy and Bayus 2018)

Average number of backers added to a project on any day



(Kuppuswamy and Bayus 2018)

Previous Works

Dynamic Research of Crowdfunding

- The U-shape dynamics allow entrepreneurs to establish effective operation strategies (Li et al. 2020)
- The success of a crowdfunding project is closely related to the promotion strategies, such as the timing of the promotion and the intensity of efforts (Lee et al. 2021)

Average number of backers added to a project on any day

Project valuation (quality $\bar{X}\beta$)	Promotion timing	
	Early (Periods 1–10)	Late (Periods 11–20)
90% quantile	0	0
75% quantile	300	925
Average	2200	8300
25% quantile	57,600	377,000
10% quantile	634,000	>10,000,000

- Early promotion is more effective
- Compared to the early promotion, the entrepreneur needs to attract more than three times the number of potential backers with later promotion to make the crowdfunding successful

(Li et al. 2020)

Note. Numbers in the cells are the additional number of potential backers required to visit the project page during the promotion period in order to bring the project up to the critical threshold for successful funding.

Previous Works

Dynamic Research of Crowdfunding

- The U-shape dynamics allow entrepreneurs to establish effective operation strategies (Li et al. 2020)
- The success of a crowdfunding project is closely related to the promotion strategies, such as the

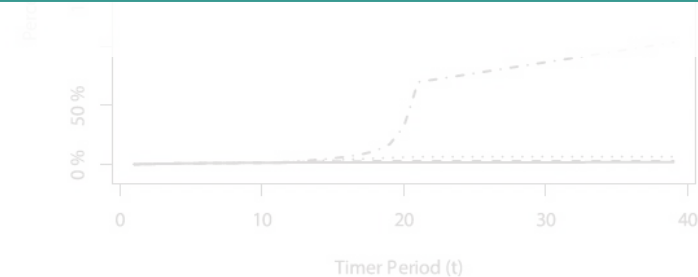
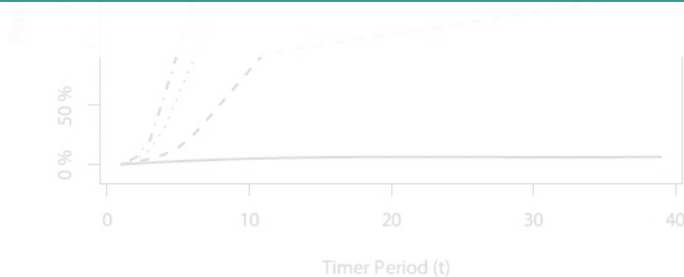
Previous studies have analyzed the crowdfunding dynamics in aggregation

(a) Early Promotion

(b) Late Promotion

However,

Each crowdfunding project may have a different degree of uncertainty
(e.g., awards, success of previous project)

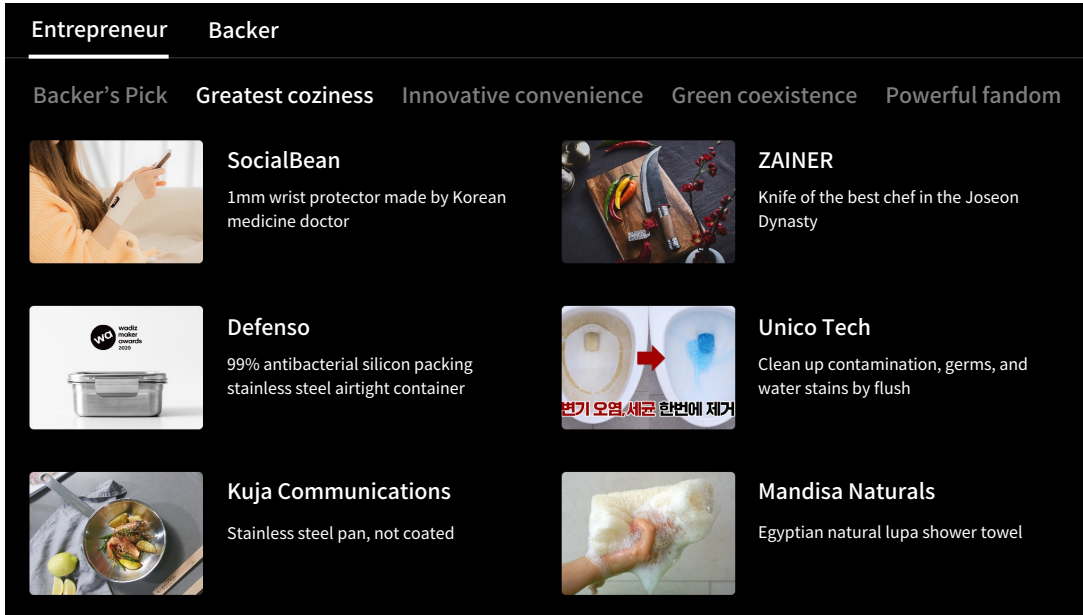


Previous Works

Awards as the Quality Signal

- The two features of the award (i.e., information content and observability) allow the award to serve as a quality signal which can reduce uncertainty in information asymmetries (Foerderer et al. 2021)
- Figures below describes the example of information content and observability

Example of Crowdfunding Platform Awards



Two Features of the Award

① Information Content (e.g., Wadiz)

Only 154 crowdfunding received an award out of 11,512 crowdfunding in 2021

② Observability (e.g., Wadiz)

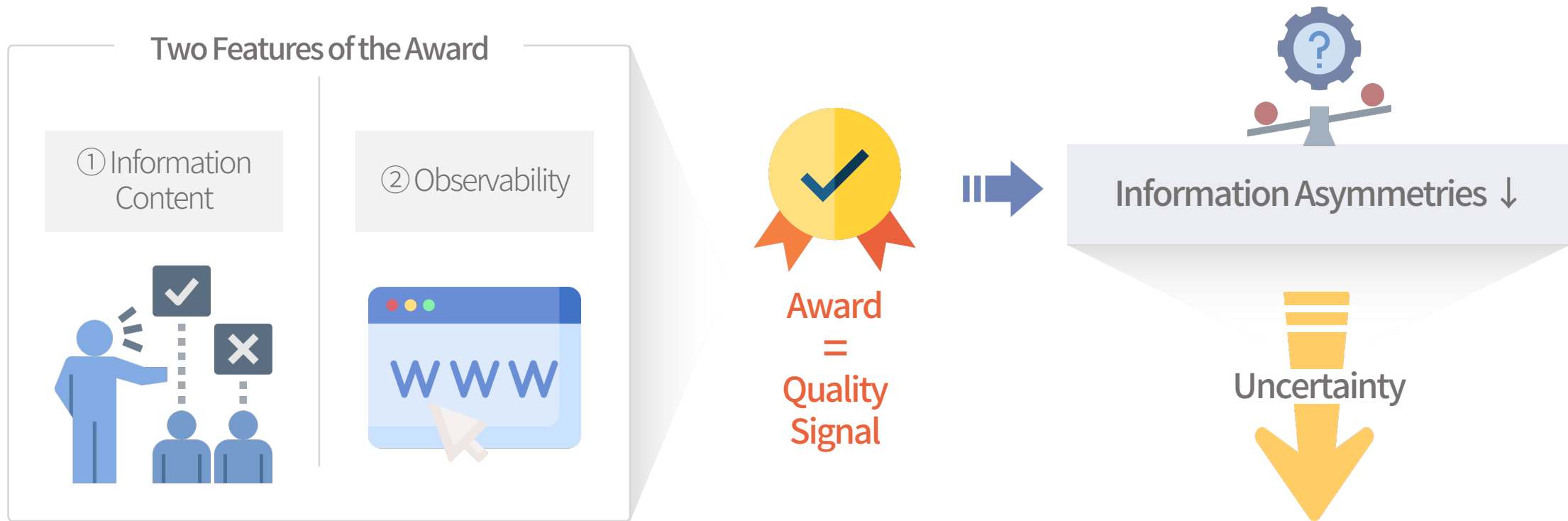
Every user can observe the award-winning entrepreneurs list on the web page

* Wadiz is a crowdfunding platform in South Korea

Previous Works

Awards as the Quality Signal

- Quality signals help to reduce information asymmetries between backers and entrepreneurs (Busenitz et al. 2005, Courtney et al. 2017, Polzin et al. 2018)



Previous Works

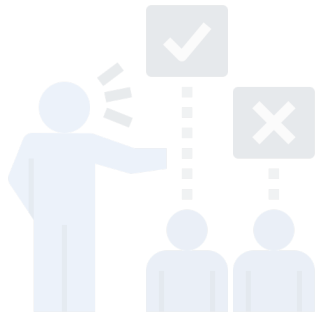
Awards as the Quality Signal

- Quality signals help to reduce information asymmetries between backers and entrepreneurs (Busenitz et al. 2005, Courtney et al. 2017, Polzin et al. 2018)

Two Features of the Award

Then, given the **mitigated uncertainty** due to **awards**,
how does the **dynamic pattern** change?

① Information Content



Award
=
Quality
Signal

Uncertainty
↓

Research Question

RQ1

Does winning an award increase the number of daily backers of subsequent projects?

RQ2

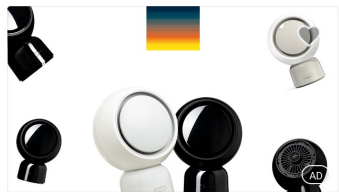
If so, how will the award effect change the number of daily backers differently in the initial, middle, and ending periods?
Does it still U-shape dynamics?

Part 3 ---

Research Design & Results

Research Context

Data Source



[Delivery before Christmas] Gift of warmth!
The hot air fan is also good looking!

tech, home appliances | Cell Mong

8165 % · KRW 40,829,100 8 days remaining



[Official] Is it custom? Win with the official
League of Legends wireless charging...

tech, home appliances | Inteco A T Co., Ltd.

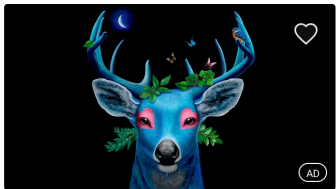
3515 % · KRW 17,577,750 23rd remaining



N company 1st place, laundry lint remover
king | First funding encore from suctio...

tech, home appliances | Momos Co., Ltd.

3078 % · KRW 15,394,000 14 days remaining



[Beauty on the verge of extinction] Koh
Sang-woo limited edition utility NFT

travel and leisure | L7 HOTELS

350 % · 3,500,000 won 7 days remaining



[Second Nation Official] The second
country is in my room! From hooded...

Character Goods | Netmarble

1719 % · KRW 17,194,200 23rd remaining



[Body lighter 1mm] A new concept
functional insole developed by a national...

Fashion, miscellaneous goods | basic balance

846 % · KRW 4,234,000 17th remaining

Platform Information

Company Name	Wadiz
Country	South Korea
Established Date	May 1, 2012
Crowdfunding Type	Reward-Based, Equity-Based

Wadiz Crowdfunding Statistics in 2021

Number of Projects	11,512
Number of Backers	551,581
Fundraising Amount	\$123,201,031

Research Context

Data

- December 23, 2020 – November 20, 2022
- 161,691 observations

Variables	Description
DailyBackers _{it}	Daily number of backers added in entrepreneur's crowdfunding project <i>i</i> at time <i>t</i>
EntrepreneurAward _i	Dummy variable; 1 if a crowdfunding project <i>i</i> is run by an award-winning entrepreneur and 0 otherwise
AfterAward _t	Dummy variable; 1 for the date after the treatment date and 0 for the date before
Cycle ^{period}	Categorical variable indicates the period of a crowdfunding cycle (initial/middle/ending)

Research Context

Data

- December 23, 2020 – November 20, 2022
- 161,691 observations

Variables	Description
EntrepreneurSatisfactionLevel	Entrepreneur's satisfaction score for past funding
EntrepreneurProjectCnt	Number of crowdfunding projects run by entrepreneurs
EntrepreneurMeanAchievementRate	Average achievement of crowdfunding projects run by entrepreneurs in 2021
EntrepreneurMeanCommunityCnt21	Average number of community posts for crowdfunding projects run by entrepreneurs in 2021
EntrepreneurMeanSupporterCnt21	Average number of supporters(number of sponsors and Facebook shares) of crowdfunding projects run by entrepreneurs in 2021
Crowdfunding Period	Duration of funding
Crowdfunding Avg Reward Price	Average price of crowdfunding rewards

Research Design

Difference-in-Differences Approach

- On December 9, 2021, our study context posted the list of award-winning entrepreneurs for 2021 on the website



Preparation for Analysis

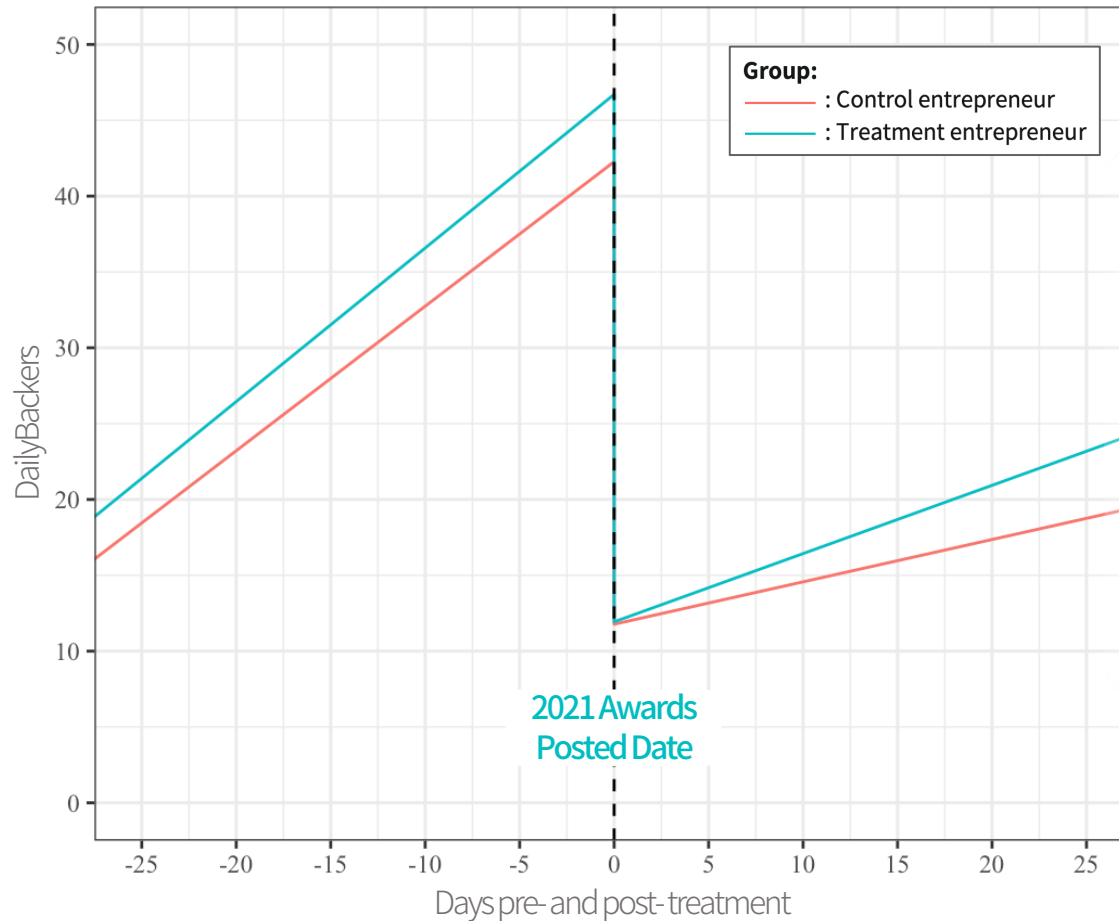
PSM Between Treatment / Control Entrepreneurs

$$\text{EntrepreneurAward}_i = \text{logit}(\text{EntrepreneurSatisfactionLevel}_i, \text{EntrepreneurProjectCnt}_i, \text{EntrepreneurMeanAchievementRate21}_i, \text{EntrepreneurMeanCommunityCnt21}_i, \text{EntrepreneurMeanSupporterCnt21}_i)$$

Matching Covariates	Before Matching				After Matching			
	Mean (Ctrl)	Mean (Trt)	t-value	p-value	Mean (Ctrl)	Mean (Trt)	t-value	p-value
EntrepreneurSatisfactionLevel	4.501	4.518	-5.078	0.000	4.507	4.501	1.2941	0.196
EntrepreneurProjectCnt	16.506	25.978	-57.160	0.000	23.939	24.162	-0.870	0.384
EntrepreneurMeanAchievementRate21	95.221	99.727	-107.530	0.000	99.987	99.681	21.529	0.000
EntrepreneurMeanCommunityCnt21	35.829	120.964	-102.640	0.000	107.781	98.856	6.876	0.000
EntrepreneurMeanSupporterCnt21	256.506	833.941	-117.520	0.000	717.865	721.092	-0.312	0.755

Preparation for Analysis

Parallel-Trend Assumption



- **X:**
Days pre- and post- treatment
- **Y:**
Daily number of backers added in entrepreneur's crowdfunding project
- **Vertical black line:**
2021.12.09 (2021 Awards Posted Date)
- **Group:**
 - : Control entrepreneur
 - : Treatment entrepreneur

DID Results

Basic DID Model (Model 1)

DV: ln(DailyBackers)		Model 1	
		Estimate	Std. Error
DID Estimates			
EntrepreneurAward		0.927	0.582
AfterAward		-0.617 ***	0.023
EntrepreneurAward x AfterAward		0.167 ***	0.032
Controls			
ln(Crowdfunding Period)		-0.147	0.014
ln(Crowdfunding Avg Reward Price)		-0.129	0.011
Entrepreneur Heterogeneity Control	Yes	# Obs	25,569
Time Heterogeneity Control	Yes	F-stats Estimate	42.57 ***

*p<0.1 **p<0.05 ***p<0.01

- **Positive effects** of winning an award
- Winning an award can increase the average number of crowdfunding project daily backers by 16.7% (about 3.7 backers)

DID Results

DID with Funding Dynamics (Model 2 – Initial: 0~25%, Middle: 25~75%, Ending: 75~100%)

DV: ln(DailyBackers)		Model 2	
		Estimate	Std. Error
DID Estimates			
EntrepreneurAward		0.211	0.540
AfterAward		-0.632 ***	0.022
Cycle ^{initial(0~25%)}		0.767 ***	0.021
Cycle ^{middle(25~75%)}		-0.233 ***	0.019
EntrepreneurAward x AfterAward		0.119 ***	0.040
EntrepreneurAward x AfterAward x Cycle ^{initial(0~25%)}		0.094 **	0.044
EntrepreneurAward x AfterAward x Cycle ^{middle(25~75%)}		0.074 *	0.039
Controls			
ln(Crowdfunding Period)		-0.197 ***	0.013
ln(Crowdfunding Avg Reward Price)		-0.132 ***	0.010
Entrepreneur Heterogeneity Control	Yes	# Obs	25,569
Time Heterogeneity Control	Yes	F-stats Estimate	63.83 ***

How award effects change over funding dynamics



- **Positive effects** of winning an award
- After the award posted date, the award-winning entrepreneur's average number of daily backers in the **initial period increases by 9.4% more** than the ending period (about 2 backers)

*p<0.1 **p<0.05 ***p<0.01

DID Results

DID with Funding Dynamics (Model 2 – Initial: 0~25%, Middle: 25~75%, Ending: 75~100%)

DV: ln(DailyBackers)	Model 2	
	Estimate	Std. Error
DID Estimates		
EntrepreneurAward	0.211	0.540
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Entrepreneur Heterogeneity Control	Yes	# Obs	25,569
Time Heterogeneity Control	Yes	F-stats Estimate	63.83 ***

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How award effects change over funding dynamics



- Moreover, after the award posted date, the award-winning entrepreneur's average number of daily backers in the **middle period increases by 7.4% more** than the ending period (about 1.6 backers)
- The award can mitigate the middle period malaise**

DID Results

DID with Funding Dynamics (Model 3 – Initial: 0~33%, Middle: 33~67%, Ending: 67~100%)

DV: ln(DailyBackers)	Model 3	
	Estimate	Std. Error
DID Estimates		
EntrepreneurAward	0.431	0.554
AfterAward	-0.624 ***	0.022
Cycle ^{initial(0~33%)}	0.608 ***	0.019
Cycle ^{middle(33~67%)}	-0.190 ***	0.020
EntrepreneurAward x AfterAward	0.115 ***	0.038
EntrepreneurAward x AfterAward x Cycle ^{initial(0~33%)}	0.097 **	0.039
EntrepreneurAward x AfterAward x Cycle ^{middle(33~67%)}	0.077 *	0.040
Controls		
ln(Crowdfunding Period)	-0.181 ***	0.014
ln(Crowdfunding Avg Reward Price)	-0.129 ***	0.010

Entrepreneur Heterogeneity Control	Yes	# Obs	25,569
Time Heterogeneity Control	Yes	F-stats Estimate	56.13 ***

*p<0.1 **p<0.05 ***p<0.01

How award effects change over funding dynamics



- **Positive effects** of winning an award
- Similar to model 2, after the award posted date, the award-winning entrepreneur's average number of daily backers in the **initial period increases by 9.7% more** than the ending period (about 2.2 backers)

DID Results

DID with Funding Dynamics (Model 3 – Initial: 0~33%, Middle: 33~67%, Ending: 67~100%)

DV: ln(DailyBackers)	Model 3	
	Estimate	Std. Error
DID Estimates		
EntrepreneurAward	0.431	0.554
AfterAward	-0.624 ***	0.022
Cycle ^{initial(0~33%)}	0.608 ***	0.019
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Controls		
ln(Crowdfunding Period)	-0.181 ***	0.014
ln(Crowdfunding Avg Reward Price)	-0.129 ***	0.010

Entrepreneur Heterogeneity Control	Yes	# Obs	25,569
Time Heterogeneity Control	Yes	F-stats Estimate	56.13 ***

*p<0.1 **p<0.05 ***p<0.01

How award effects change over funding dynamics



- Also, after the award posted date, the award-winning entrepreneur's average number of daily backers in the **middle period increases by 7.7% more** than the ending period (about 1.7 backers)
- The award can mitigate the 'stuck in the middle' problem**

Part 4

Conclusion

Main Findings

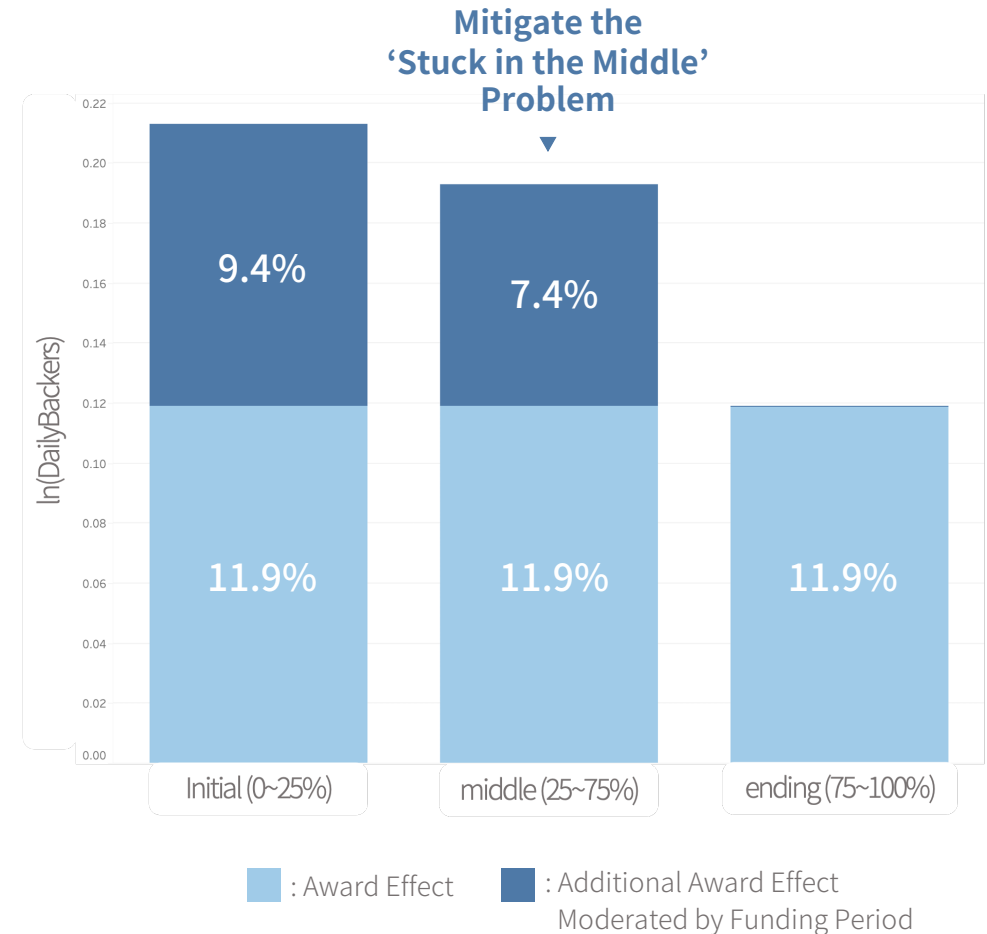
Award Effects

- Winning an award increase the average number of crowdfunding project daily backers by more than 15%
- The award serves as a quality signal

Heterogeneity in the Funding Dynamics

- The award effect varies depending on the funding dynamics
 - The award effect increases the average number of daily backers by 9% or more in the initial period and 7% or more in the middle period than the ending period
- The award can mitigate the 'stuck in the middle' problem

Award Effect by Funding Period



Implications

Academical Implications

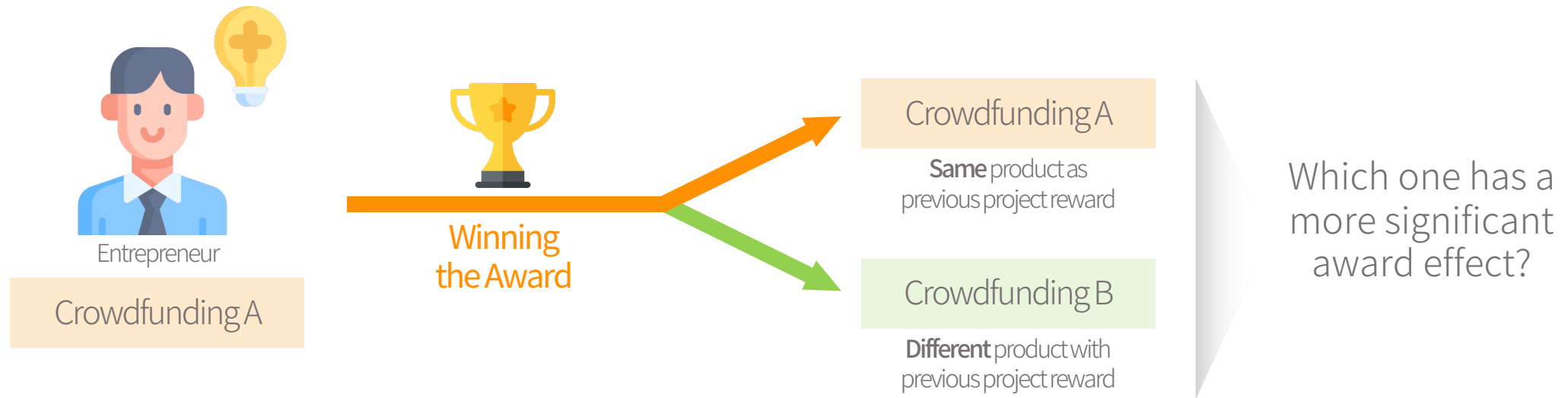
- We take a step further from prior literature by analyzing the dynamics of crowdfunding backers' behaviors over the crowdfunding cycle under less uncertain situations
- The dynamics of award-winning entrepreneur's crowdfunding become the strong L-shape

Practical Implications

- The award can mitigate the middle period malaise
- Given the reduced uncertainty of the backers, entrepreneurs should take a different strategy from that of normal crowdfunding (e.g., change the promotion timing)

Future Plan

1. Analyze the difference in award effect depending on whether an entrepreneur runs funding with the same product or with a different product after winning the award



2. Analyze the difference in award effect by award types

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Thank you

Is It Still a U-shape?

Empirical Study on the Dynamics of Reward-Based Crowdfunding under Mitigated Uncertainty

Appendix

Appendix

Descriptive Statistics of Collected Data

	mean	std	min	25%	50%	75%	max
DailyBackers	11.192	40.082	0.000	1.000	3.000	8.000	3365.000
EntrepreneurSatisfactionLevel	4.509	0.479	1.000	4.000	4.500	5.000	5.000
EntrepreneurProjectCnt	17.852	19.870	2.000	5.000	10.000	22.000	117.000
EntrepreneurMeanAchievementRate21	2657.859	5305.310	0.000	340.250	934.500	2620.500	83136.000
EntrepreneurMeanCommunityCnt21	43.240	73.620	0.000	7.000	19.333	54.000	1187.000
EntrepreneurMeanSupporterCnt21	274.489	349.386	2.500	67.063	153.800	336.600	3682.250
Crowdfunding Period	19.801	8.533	1.000	14.000	19.000	24.000	236.000
Crowdfunding Avg Reward Price	96.55	160.54	2.87	33.12	53.54	98.76	5911.78

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